



Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

AJAX PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Web Technologies

TOTAL: 10 QUESTIONS

Q1 What is AJAX and what problem in Web Technologies does it solve? **Introduction**

Q6 How does AJAX handle reliability and high availability? **Observability**

Q2 What are the core components or concepts in AJAX? **Architecture**

Q7 What observability does AJAX provide out of the box? **Lifecycle**

Q3 How does AJAX compare to its main alternatives? **Configuration**

Q8 What are common performance bottlenecks in AJAX? **Compliance**

Q4 How do you get started with AJAX for a new project? **Hardening**

Q9 What security best practices apply when running AJAX? **Testing**

Q5 What configuration options most affect AJAX's behavior in production? **SSL / IDP**

Q10 What mistakes do teams commonly make when adopting AJAX? **Tuning**



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 AJAX is a tool or platform that

Mitigation

AJAX is a tool or platform that automates or simplifies a specific concern in the Web Technologies domain, reducing manual effort and operational complexity for engineering teams.

A6 AJAX provides HA through leader election, replication, or stateless horizontal scaling.

Availability

AJAX provides HA through leader election, replication, or stateless horizontal scaling. Deploy at least two instances across availability zones and configure health checks for automatic failover.

A2 AJAX has key building blocks that work

Primitives

AJAX has key building blocks that work together: a configuration layer defines intent, a runtime layer enforces it, and an API or CLI layer allows operators to manage and observe the system.

A7 AJAX exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces.

Information

AJAX exposes metrics (Prometheus endpoint or built-in dashboard), structured logs, and optionally distributed traces. Define SLOs on the key metrics and alert before users are affected.

A3 AJAX trades off simplicity against feature richness

Hardening

AJAX trades off simplicity against feature richness differently than its alternatives. Choose AJAX when its specific strengths performance, ecosystem, or ease of use align with your team's primary constraints.

A8 Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches.

Performance

Performance bottlenecks typically stem from misconfigured connection pools, large payload sizes, insufficient worker threads, or missing caches. Profile with built-in diagnostics before optimizing.

A4 Install AJAX via its package manager or binary release.

Gotchas

Install AJAX via its package manager or binary release. Follow the quickstart to initialize a project, configure the minimal required settings, and run the default command to verify the setup works end-to-end.

A9 Run AJAX with the principle of least privilege

Assessment

Run AJAX with the principle of least privilege, enable TLS for all communication, use a secrets manager for credentials, and keep the software patched to the latest stable release.

A5 Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels.

Configuration

Production-critical settings typically include concurrency limits, timeout values, retry policies, resource limits, and logging levels. Review the official production hardening guide before deploying.

A10 Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.

Configuration

Common pitfalls include skipping the documentation and learning the API by trial and error, using default configurations in production, lacking a runbook for operational issues, and not testing failover scenarios.